Network Meta-Analysis: Introduction

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Introduction

Network meta-analysis (NMA), or mixed-treatment comparison (MTC), pools evidence across studies comparing different pairs of treatments to estimate a coherent set of relative effects. It enables comparisons for which no direct head-to-head trials exist, provided indirect paths in the evidence network connect the treatments.

Prerequisites

Pairwise meta-analysis; graph structure; consistency concept.

Theory

A network has nodes (treatments) and edges (pairwise comparisons with at least one study). Direct evidence = studies comparing treatments directly; indirect evidence = inferred via a common comparator. The pooled estimate combines both under an assumption of **consistency** (direct = indirect estimates).

Implementation: frequentist (`netmeta`) or Bayesian (`gemtc`, `BUGSnet`).

Assumptions

Transitivity: studies in the network are sufficiently similar that indirect comparisons are valid. **Consistency**: direct and indirect estimates of the same contrast agree within sampling error.

R Implementation

```r
library(netmeta)

# Example dataset
set.seed(2026)
net_df <- data.frame(
  study = c("S1", "S1", "S2", "S2", "S3", "S3", "S4", "S4", "S5", "S5"),
  treat = c("A", "B", "A", "C", "B", "C", "A", "D", "C", "D"),
  TE    = c(0, -0.3, 0, -0.5, 0.4, 0.2, 0, -0.2, 0, 0.3),
  seTE  = c(0.2, 0.2, 0.25, 0.25, 0.3, 0.3, 0.2, 0.2, 0.22, 0.22)
)
# Rearrange to pairs
p_df <- pairwise(treat = treat, TE = TE, seTE = seTE,
                 studlab = study, data = net_df)

net <- netmeta(p_df, common = FALSE, reference = "A")
summary(net)
netgraph(net)
```

Output & Results

A network graph, pooled treatment effects relative to the reference, and overall I^2.

Interpretation

"Network meta-analysis of 5 studies covering 4 treatments (A reference, B, C, D) yielded pooled SMDs: B vs A -0.22 (-0.41, -0.03); C vs A -0.35 (-0.58, -0.12); D vs A 0.05 (-0.15, 0.25). Transitivity was plausible; consistency was not violated."

Practical Tips

- Always produce a network graph first; disconnected components cannot be compared.
- Check transitivity via study characteristics (populations, doses, outcomes).
- Use SUCRA to rank treatments by their probability of being best.
- Report PRISMA-NMA checklist; network meta-analyses are complex and demand transparency.
- Bayesian NMA via `gemtc` is more flexible for hierarchical models and missing-arm data.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis